

GEONEX - GEO HORIZON'26

GEO SYNC - MOBILE APPLICATION DEVELOPMENT

BRIDGING THE GAP BETWEEN THE OFFICE MAP AND THE GROUND REALITY

OFFLINE-FIRST • GEO FENCED • AUTO - SYNC • 360 ° PANORAMIC



TEAM MEMBERS

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PROBLEM STATEMENT

Field workers carry the most critical data — but they're still collecting it on paper, in places with no internet, with zero way to prove they were even there.



WHY ARE WE BUILDING THIS?

Every government department, infrastructure company, forest officer, and disaster relief team faces this exact problem today, right now, in India.

We are building this because:



70% of India's survey zones have poor or zero connectivity — cloud-first tools are useless here

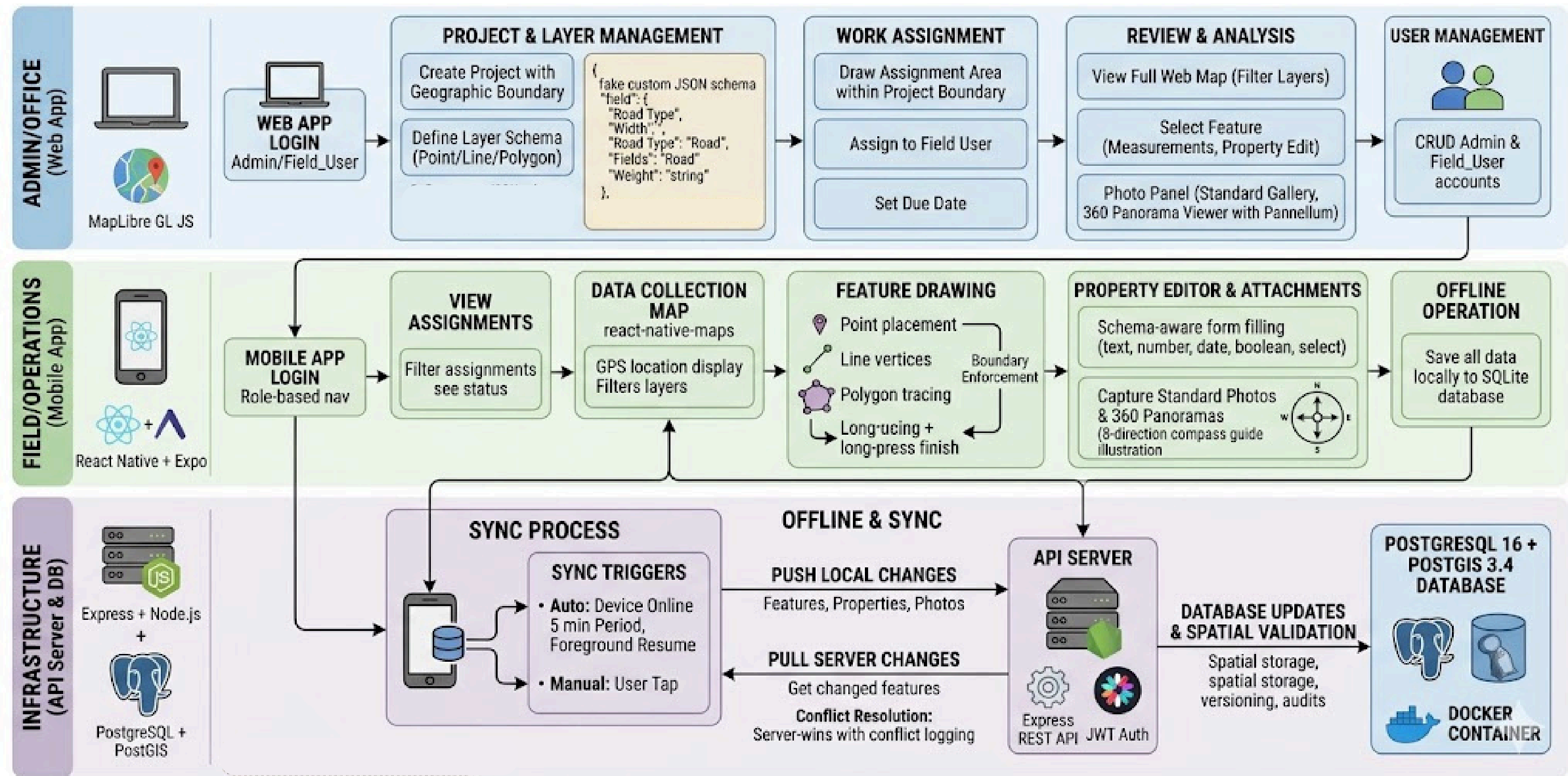


ArcGIS & commercial GIS tools cost ₹5–20 lakhs/year — open source is not just better, it's necessary

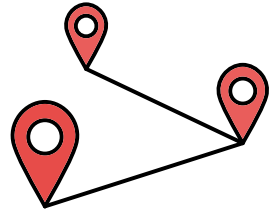


Fake field reports cost the government crores — geofenced photo proof solves this permanently

OUR PROJECT - ARCHITECTURE



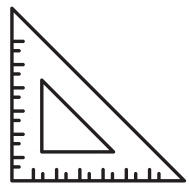
MOBILE APPLICATION



Live GPS on Map - Real-time GPS position shown as overlay. Always know where you are.



Draw Features - Tap points, trace lines, close polygons. Full GIS drawing on mobile.



Auto Measurements - Area, perimeter, length auto-calculated when you select any feature.



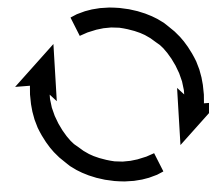
Geofencing - Cannot collect data outside assigned zone — client + server enforced.



360° Panorama - 8-direction compass-guided photo capture. Immersive field documentation.



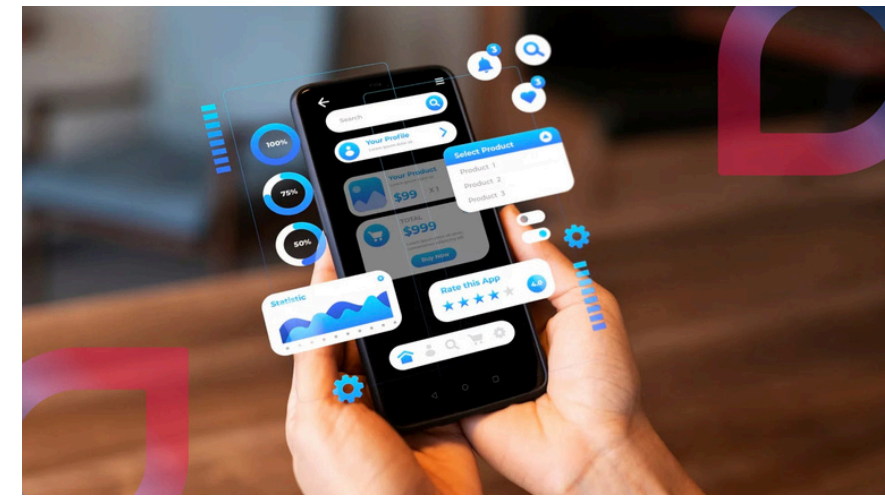
Offline Mode - Full functionality with zero internet. SQLite stores everything locally.



Auto-Sync - 4 triggers: connectivity, foreground, periodic 5min, and manual sync.



Role-Based UI - Admin sees full dashboard. Field user sees assignments only.



OFFLINE FIRST - ARCHITECTURE



NO INTERNET (Field)

✓ View assigned zones on map

✓ Draw points, lines, polygons

✓ Fill attribute forms

✓ Capture photos + 360°
panoramas

✓ Data saved to SQLite locally

✓ Marked as 'pending sync'



BACK ONLINE (Sync)



Connectivity Restored

NetInfo detects, syncs immediately



App Foreground

AppState listener triggers sync



Every 5 Minutes

Periodic check if pending exist



Manual Sync

User taps " Sync " button
to sync data to cloud

REAL WORLD APPLICATIONS



**Illegal logging
detection, wildlife &
fire zone mapping**



**Vaccination camp
mapping, sanitation
zone surveys**



**Construction milestone
tracking, defect
reporting**

**Farm boundary
mapping, soil surveys,
crop health, Loan &
subsidies**



**Road, drain & streetlight
asset mapping in urban
zones**

FUTURE SCOPE & VISION

⚡ SHORT TERMS

🔔 Push notifications for deadlines Using Redis Celery



🛰️ Satellite imagery basemap integration



🚀 LONG TERM (Vision)



To present completely user friendly experience to Naivee users



🤖 AI-powered Automatic feature extraction with area and perimeter estimation from 360° imagery



🌐 Any organization ISRO Bhuvan / NRSC portal integration/Cadastral Dept

THANK YOU !